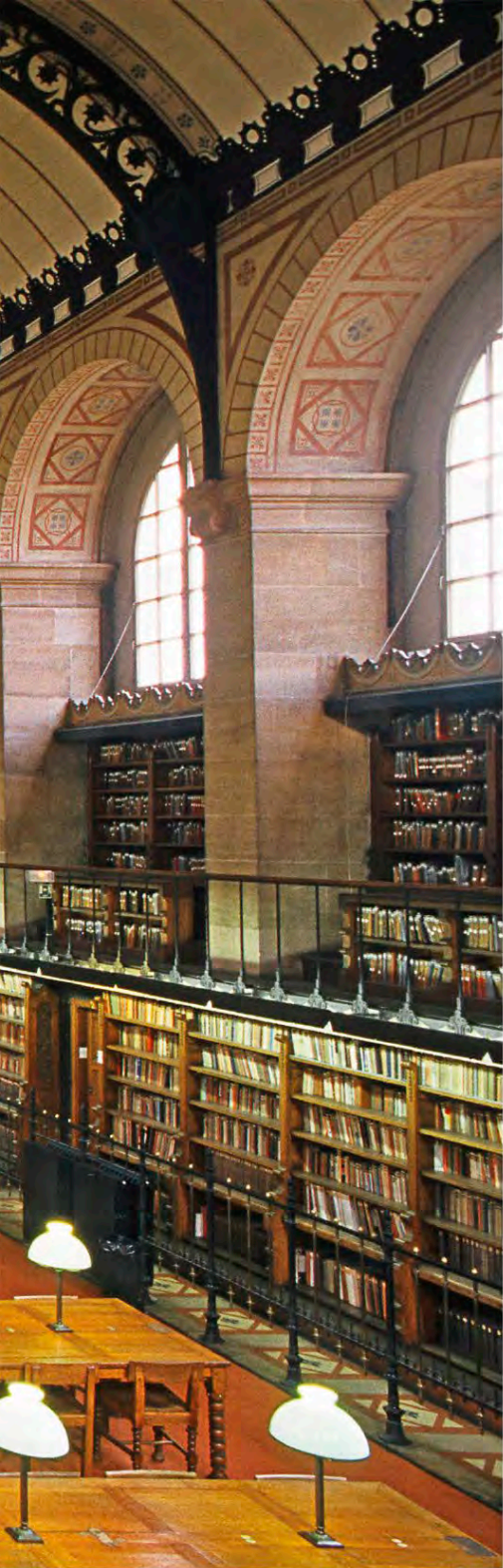




Crystal Palace

Location London, England

Joseph Paxton's Crystal Palace, as it was instantly dubbed, was the single most important building of the first phase of the Industrial Revolution of the 19th century. It was the setting for the Great Exhibition of 1851, the brainchild of Prince Albert, intended to showcase Great Britain's myriad technological advances. This called for an exceptional setting, and it got it. The Crystal Palace

was 1,851 ft (564 m) long and rose to 128 ft (39 m). Constructed entirely of iron and glass, it was prefabricated and erected on site in Hyde Park. Its design was simple: a central flat-roofed aisle flanked by two lower flat-roofed aisles. At its center, a vaulted transept large enough to contain mature trees cut across them. It was a dazzling demonstration of Britain's industrial prowess. Relocated to Penge Common, South London, after the exhibition, it was destroyed by fire in 1936.

Lutheran Cathedral

Location Helsinki, Finland

Helsinki's bold Neo-Classical cathedral dominates the heart of the city. It was designed by a German, Carl Ludvig Engel, but finished, in 1852, by a second, Ernst Lohrmann, after Engel's death in 1840. Stark and simple, it comprises a central block with four identical six-columned, pedimented porticos on each facade. It sits on what amounts to a gigantic granite podium approached by steps. Above the central block towers a hefty green dome on an elongated, two-story drum. Four smaller drums, added by Lohrmann, are at the roof corners. The grandeur of the building is enhanced by its setting in Engel's spaciouly harmonious Senate Square. The dazzlingly white, sparse interior is flooded with light.

St. Isaac's Cathedral

Location St. Petersburg, Russia

Although it is on a vastly greater scale than Helsinki's Lutheran Cathedral (see above), St. Isaac's Cathedral is essentially the same in plan: a Greek cross with four identical porticoed facades having pediments borne by giant columns. A dome, flanked by four smaller domes, dominates the

structure. However, its execution could hardly be more different from the Lutheran Cathedral. Apart from its impressive size—the building can hold 14,000 people—this comes down to materials. St. Isaac's, the work of French architect August Montferrand, completed in 1858, is ostentatious. Its domes are gilded, and its columns are made of red granite. Forty-eight bronzelike groups of sculptures stud the roof lines. The interior is more lavish still and glints with colored marble and semiprecious stones.

Clifton Suspension Bridge

Location Avon Gorge, England

There is no structure that more emphatically proclaims the technological advances of mid-Victorian Britain than the Clifton Suspension Bridge. Construction began in 1831, but the bridge was not completed until 1864. It was the engineering wonder of its day. Measuring 1,350 ft (412 m) long, it dramatically spanned the rocky Avon Gorge 250 ft (76 m) below. Its architectural

▼ THE IMPOSING TOWERS OF THE CLIFTON SUSPENSION BRIDGE



pretensions are made clear by the Egyptian motifs on the two vast splay-footed towers that support it. But it is as an example of engineering daring, in which Brunel excelled, that the bridge impresses. Although it was widely presumed that the bridge would collapse, it remains sturdily intact.

Turbine Building, Menier Chocolate Factory

Location Noisiel-sur-Marne, France

The Turbine Building at the Menier Chocolate Factory, east of Paris, is a purely functional building used to mill cocoa beans. Positioned on the River Marne, the mill's power was supplied by three water-driven turbines. It was designed by Jules Saulnier and completed in 1874. Its importance lies in the fact that although externally it is made of pale yellow bricks, its internal iron frame is tied together by diagonal bracing on the outside of the building. In other words, the bricks have no structural role at all. However, the brickwork is highly decorative, with a range of motifs based on flowers and cocoa beans. The impact is appealingly playful.